

A Genuine Noncommutative Hartman–Wintner LIL Was Obtained in 2026

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Status: literature already answered (affirmative answer to the source’s existence question; no new theorem is claimed here).

Original question

Section 4 of Zeng’s paper asks whether a “genuine” noncommutative Hartman–Wintner law of the iterated logarithm is possible, where genuine means that the theorem does not assume Kolmogorov’s pointwise small-increment condition

$$\|d_n\|_\infty \leq \alpha_n s_n / u_n, \quad \alpha_n \longrightarrow 0.$$

The source points to Stout’s classical theorem for stationary-ergodic martingale differences and leaves the noncommutative possibility open [1] (source PDF page 8).

Explicit later answer

Panja, Ricard, and Saha explicitly cite Zeng and prove optimal noncommutative LILs in arXiv:2509.22037v2 [2].

First, their Theorem 3.4 (supporting PDF page 12) is a genuine martingale version obtained by cutting large differences. Instead of the pointwise small-increment assumption, it imposes a conditional Lindeberg-type tail summability condition: for some $K_n \rightarrow 0$ and $\alpha > 0$,

$$\left\| \sum_{n \geq 1} \frac{u_n^\alpha}{s_n^2} E_{n-1} \left[d_n^2 \mathbf{1}_{(s_n K_n / u_n, \infty)}(|d_n|) \right] \right\| < \infty.$$

It concludes the optimal almost-uniform upper bound

$$\limsup_{n \rightarrow \infty} \frac{x_n}{s_n u_n} \leq \sqrt{2} \text{ a.u.}$$

Second, Theorem 4.4 (supporting PDF page 14) removes the small-increment condition completely for independent identically distributed data. If (y_i) are independent, identically distributed, mean-zero noncommutative random variables with $\|y_i\|_2 \leq 1$, then, writing $x_n = \sum_{i=1}^n y_i$ and $u_n = \sqrt{\max\{1, \log \log n\}}$,

$$\limsup_{n \rightarrow \infty} \frac{x_n}{\sqrt{n} u_n} \leq \sqrt{2} \text{ a.u.}$$

Independent identically distributed differences are stationary and ergodic. Thus the source’s question whether a genuine noncommutative Hartman–Wintner theorem is *possible* has an affirmative answer.

Scope

The later paper proves an optimal genuine martingale theorem under a tail condition and the full IID L_2 Hartman–Wintner theorem. It does not claim the strongest conceivable statement for every dependent stationary-ergodic noncommutative martingale-difference sequence. The source’s open-ended possibility question is answered, while that broader universal extension should not be inferred.

The later authors also identify a slight gap in the exponential-inequality application used for Zeng’s original constant-2 result; their refined argument both corrects the route and obtains the classical sharp constant.

References

- [1] Q. Zeng, *Kolmogorov’s law of the iterated logarithm for noncommutative martingales*, Ann. Inst. H. Poincaré Probab. Statist. **51** (2015), 1124–1130; arXiv:1212.1504v3.
- [2] S. Panja, É. Ricard, and D. Saha, *Non-commutative Law of iterated logarithm*, J. Funct. Anal. (2026), article 111477; arXiv:2509.22037v2.